

GATTON, Australia

WMO: 945620

Lat: 27.5436S

Lon: 152.3375E

Elev: 89

StdP: 100.26

Time Zone: 10.00 (E10)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.5	5.0	-4.6	2.6	17.0	-2.7	3.0	15.0	13.6	18.5	10.9	15.9	2.6	260	0.386

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.1	36.2	21.8	34.3	21.7	32.7	21.3	24.4	31.6	23.7	30.4	23.1	29.3	4.7	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.6	17.5	26.8	22.0	16.8	26.0	21.4	16.2	25.7	74.2	32.0	71.4	30.4	69.1	29.6	27.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.9	8.2	7.2	DB	0.0	40.2	1.1	2.3	-0.8	41.8	-1.5	43.2	-2.1	44.5	-3.0	46.2	(4)
			WB	-1.0	25.7	1.1	0.9	-1.8	26.4	-2.5	26.9	-3.1	27.4	-3.9	28.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.2	25.8	25.1	23.5	20.4	17.0	14.7	13.9	14.9	18.1	20.9	23.3	24.7
	DBStd	4.78	2.26	2.34	2.07	1.96	2.09	2.12	2.11	2.21	2.59	2.76	2.79	2.67
	HDD10.0	1	0	0	0	0	0	1	1	0	0	0	0	0
	HDD18.3	457	0	0	0	4	52	109	138	111	34	8	1	0
	CDD10.0	3709	488	422	418	313	218	142	122	151	244	337	400	455
	CDD18.3	1124	230	188	159	67	11	1	1	3	28	87	151	197
	CDH23.3	10204	2118	1676	1209	533	128	11	18	97	432	866	1343	1773
	CDH26.7	3982	952	703	396	104	9	1	1	19	136	321	558	782
Wind	WSAvg	3.2	3.3	3.3	3.1	2.8	3.0	3.2	3.3	3.4	3.2	3.2	3.3	3.2
Precipitation	PrecAvg	796	116	98	73	52	59	37	42	27	31	67	90	102
	PrecMax	1087	453	300	219	324	444	320	306	91	89	237	219	265
	PrecMin	355	16	14	3	0	0	0	0	0	0	9	6	22
	PrecStd	192	83	68	52	63	76	55	57	22	26	49	55	52
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.1	38.1	34.9	30.7	27.9	24.7	25.5	29.0	33.3	35.7	37.2	38.1
		MCWB	23.2	23.7	22.4	19.8	17.4	16.2	15.4	16.3	17.7	18.7	20.4	21.6
	2%	DB	35.4	34.6	32.1	29.1	26.1	23.0	23.2	25.8	30.4	32.4	34.3	35.2
		MCWB	22.6	23.2	21.9	19.3	16.7	15.6	14.4	14.9	17.0	18.6	20.5	22.0
	5%	DB	33.4	32.7	30.4	27.8	24.8	21.7	21.6	23.7	27.8	30.1	32.0	32.8
		MCWB	22.2	22.4	21.1	18.9	16.1	14.8	14.0	14.0	16.3	18.3	20.2	21.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	10%	DB	31.6	30.8	28.9	26.5	23.4	20.4	20.1	22.1	25.6	28.1	30.0	30.8
		MCWB	22.1	21.7	20.8	18.5	15.7	14.3	13.2	13.3	15.9	17.8	19.8	21.2
	0.4%	WB	25.4	25.6	24.2	22.1	19.7	18.2	18.7	18.1	20.1	22.3	23.6	24.5
		MCDB	32.7	32.7	30.0	28.0	25.3	21.3	22.1	24.9	27.3	29.6	31.7	32.9
	2%	WB	24.5	24.4	23.2	21.0	18.5	17.1	16.8	16.7	18.9	20.9	22.5	23.6
		MCDB	31.3	31.7	28.8	26.3	23.4	20.8	19.9	22.5	25.6	27.5	29.8	31.1
Mean Daily Temperature Range	5% DB	WB	23.7	23.6	22.6	20.1	17.5	16.3	15.5	15.6	17.9	19.9	21.6	23.0
		MCDB	30.2	29.9	27.8	25.3	22.3	19.8	19.2	21.1	24.7	26.3	28.8	29.7
	5% WB	WB	23.1	22.8	22.0	19.3	16.7	15.5	14.3	14.5	17.0	19.0	20.8	22.3
		MCDB	28.7	28.6	26.4	24.7	21.5	19.0	18.6	20.2	23.4	25.1	27.2	28.1
Clear-Sky Solar Irradiance	taub	0.386	0.373	0.353	0.333	0.312	0.304	0.295	0.308	0.349	0.373	0.382	0.394	
	taud	2.481	2.526	2.576	2.610	2.615	2.635	2.644	2.586	2.467	2.433	2.451	2.442	
	Ebn at Noon	957	952	942	915	893	879	904	929	931	941	954	951	
	Edh at Noon	117	109	99	87	78	73	75	87	108	119	120	123	
All-Sky Solar Radiation	RadAvg	6.55	5.99	5.20	4.55	3.79	3.14	3.62	4.53	5.53	6.13	6.64	6.60	
	RadStd	0.56	0.56	0.31	0.23	0.21	0.23	0.28	0.34	0.45	0.60	0.64	0.69	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (27 neighbors)		+0.41	+0.52	N/A	N/A	N/A	N/A	N/A	-67	+186	+112

Nomenclature: See separate page